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# Active Lifelong Anomaly Detection with Experience Replay

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***Abstract*—Anomaly detection tools present the potential to enhance defense policies and protection against different types of threats, supporting public safety and national security. Lifelong anomaly detection showcases new and challenging scenarios in which models are challenged to automatically adapt to changing conditions without forgetting past knowledge. However, the presence of anomalies in incoming data may significantly impact the robustness of models in such scenarios. Although active learning strategies could be an asset to increase model longevity and robustness, they have never been explored in this context. In this paper, we propose an active lifelong anomaly detection framework for class-incremental scenarios that supports any memory-based experience replay method, any query strategy, and any anomaly detection model. While experience replay allows models to consolidate past knowledge and simultaneously adapt to new knowledge, an active learning module reduces the number of anomalies memorized in the replay buffer. We propose two strategies that automatically identify and remove additional data points that are likely to be anomalies based on the oracle's feedback. Our experiments on popular host-based and network-based intrusion detection datasets show that our framework can improve the anomaly detection performance of models under low labeling budget constraints.**

***Index Terms*—anomaly detection, intrusion detection, lifelong learning, active learning, experience replay**

## I. Introduction

Anomaly detection models represent important decision support tools in real-world domains and are an essential component in many cybersecurity applications. Relevant examples include detecting intrusions and attacks in network traffic and in smart grids, as well as detecting cyber threats such as distributed denial-of-service (DDoS) attacks, spamming, and attacks directed to mobile and IoT devices [1]–[4]. In this context, anomaly detection models have the potential to mitigate threats and protect both private companies and national security, fostering public safety at the environmental and at the infrastructure level.

Despite the progress made in recent years [5], [6] one crucial drawback of conventional anomaly detection models is that, by focusing on adaptation to rapidly changing environments, i.e. high plasticity, they are subject to forgetting past knowledge, i.e. limited stability. This issue results in the inability to handle recurring concepts that are common in real-world applications. Lifelong learning is an emerging machine learning paradigm that attempts to find a proper stability-plasticity trade-off involving cross-disciplinary knowledge in the fields of biology, neuroscience, and computer science [7]. Besides the fact that most of the research works in lifelong learning are directed to image classification and reinforcement learning [7], [8], lifelong learning for anomaly detection problems has the potential to yield sophisticated models that are capable of detecting anomalies while adapting to changing environments and avoiding forgetting knowledge acquired in the past [9]–[11].

A different thread of research is that of active learning for anomaly detection, which addresses the challenge of labeled background data availability in real-world settings. Since anomalies are likely to be present in raw background data, memory-based lifelong learning approaches such as experience replay may be significantly affected, since they are likely to incorporate anomalies in replay buffers used for model updates. In turn, models may be less robust, yielding a sub-optimal anomaly detection performance. Therefore, active learning in the context of lifelong anomaly detection has the potential to overcome this issue. However, to the best of our knowledge, there are still no studies that focus on combining the peculiar features of lifelong anomaly detection and active learning.

In this paper, we propose an active lifelong anomaly detection framework for class-incremental scenarios which supports any memory-based experience replay method, and any query strategy. While experience replay is adopted to consolidate past knowledge and simultaneously allow for adaptation to new knowledge, an active learning module aims to ensure that the consolidated past knowledge presents none or a limited

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number of anomalies.

We devise two strategies to select a limited number of data samples as queries for the oracle. In addition, the strategies exploit queries labeled as anomalies to further explore the replay buffer in order to identify and remove additional data points that are likely to be anomalies. By doing so, they aim to remove a larger number of anomalies than the budget (i.e. the number of available queries or data points to be labelled by the oracle), thus reducing contamination in the replay buffer, which in turn yields more robust models. Our framework is model-agnostic, since it can be applied to any off-the-shelf semi-supervised or unsupervised anomaly detection model. In this work, we put particular focus on intrusion detection, since it is an essential asset to enhance defense policies, supporting public safety and national security. We evaluate our approach on three common host-based and network-based intrusion detection datasets. Experimental results show that our approach is capable of improving the anomaly detection performance of frequently adopted anomaly detection models under low labeling budget constraints.

The paper is structured as follows. Section II summarizes related works and describes lifelong learning scenarios. Section III describes our framework and proposed strategies. Section IV describes the experimental settings, baseline strategies and scenarios used in our study. Section V presents and discusses the results. Finally, Section VI concludes the paper summarizing the key results achieved and outlining ideas for future work.

## II. Background

In this section, we describe works in the literature pertaining to our study. We first describe commonly adopted lifelong learning scenarios and provide a categorization of lifelong learning methods. We proceed with an overview on active learning methods. Finally, we focus on anomaly detection methods and models that incorporate an active learning component.

### A. Lifelong learning scenarios

A lifelong learning setting usually consists of a continuous process in which a sequence of tasks is presented to the learning model [7], [12]. The goal of the learner is not only to adapt to the changing environment, but also to keep the knowledge acquired from the previously learned tasks. In this setting, the model is systematically evaluated on all tasks, to assess how well it performs in terms of adaptation and knowledge preservation. One popular scenario in image classification is the Split-MNIST dataset [13], which splits 10 classes of handwritten digit images in 5 binary tasks containing 2 digits each.

Scenarios are often categorized based on the availability of two key characteristics: task labels and task boundaries[1]. The availability of task boundaries implies that the model is aware during the learning phase that a new task occurs, and that new data comes from a new task, allowing the model to incorporate data and learn the new task. In addition to knowledge on task boundaries, the model may also be informed about which task is currently being presented (task label) during both the learning and the inference phases. Task-incremental scenarios provide the model with both task labels and task boundaries, which means that the model is aware that a new task is presented and knows which task is currently under evaluation. On the other hand, class-incremental scenarios provide the model with only task boundaries, which means that the model is aware of a task transition, but has no knowledge about which task is currently presented [12]. Similarly, class-incremental scenarios can be observed in recent lifelong anomaly detection works [10]. The main difference between anomaly detection and image classification in this context is that, in anomaly detection, a task provides a new characteristic of the normal class, instead of new classes that need to be classified. The goal of a lifelong anomaly detector is to adapt to the new behavior of the normal class, while still yielding satisfactory results on the previous characteristics of the normal class.

[1]Although more distinctive properties of lifelong learning scenarios can be identified, our discussion focuses on these two properties since they are the most pertinent for our study.

### B. Lifelong learning methods

Lifelong learning methods are focused on improving the performance of neural networks in specific lifelong learning scenarios. We start with a general characterization of methods into three categories [7] and then we focus on the most related one for this study, i.e. experience replay.

The first category includes regularization-based methods that work by introducing constraints on weight updates during the incremental training of neural networks. One popular strategy is preventing the model from updating weights learned on previously trained tasks [14]. The second category focuses on dynamic architectures, in which the architecture of the network changes during the learning process. A notable example is expanding the network with new neurons or layers [15]. The third category, which is the most relevant for the research conducted in our study, includes replay-based methods, which recall prior data from previously seen tasks during the model update, to make sure that their characteristics are included in the updated model. There are two main categories of replay-based methods [7]. The first one focuses on selecting and storing the original data samples corresponding to learned tasks in the memory buffer. The stored data samples are then provided to the model in a so-called replay buffer. The second category is called Generative Experience Replay [16], as it leverages generative models to generate data reflecting the previously learned characteristics each time the model update process is executed. In our framework, we adopt a state-of-the-art combination of balanced replay buffers and reservoir sampling described and analyzed in [17]. While balanced replay buffer ensures that classes are equally (or almost equally) represented, reservoir sampling [18] decides which samples from each class should be remembered.

Most of the recent lifelong learning research is oriented to image classification and reinforcement learning. However,

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lifelong anomaly detection methods are gaining momentum. One example of recent works is ARCADe, a meta-learning method that aims to identify the most appropriate parameter values for every single task in one-class image classification [10]. Another notable example that shows the real-life application of lifelong anomaly detection is a transfer learning method designed for detecting anomalies in surveillance videos [11]. Finally, a recent lifelong anomaly detection work closely related to online anomaly detection is CDPGA, which leverages change point detection and a dynamic architecture, presenting improved performance on new tasks [9]. Emerging studies confirm the usefulness of non-parametric change point detection in real-world applications [19], also to identify emerging tasks and recurrent tasks in time series data [20].

### C. Active learning

The ability to gather enough labeled data to support training machine learning models represents an obstacle in many real-world domains. Labeled data is expensive to acquire and frequently relies on human labor. On the other hand, unlabeled data is abundant in many fields. The general challenge addressed by active learning methods is that of improving predictive models in scenarios with limited availability of labels, acquiring labels from an oracle for a limited portion of data samples. Active learning methods are composed by two primary components: an *oracle* that provides responses to a query, and a *query system*, which poses queries to the oracle [21]. Selecting the most appropriate data samples to be provided to the oracle is the most important capability of active learning methods. While methods based on *membership query synthesis* actively synthesize samples from the entire data space, *sequential sampling* methods draw samples from the underlying data distribution. The vast majority of methods in the literature, however, adopt a *pool-based sampling* approach, where a set of examples are labeled at a given time during the active learning process.

Querying strategies can be grouped in four main categories. *Heterogeneity-based models* aim at sampling from heterogeneous and uncertain regions of the data space. Examples include uncertainty sampling [22], query-by-committee [23], and expected model change [24]. *Performance-based models* tend to focus on the most unknown, noisy, or unrepresentative regions of the data space. One prominent example is expected error reduction [25], which minimizes the expected label uncertainty of data samples that are not yet labeled. *Representativeness-based models* tend to avoid querying queries on unrepresentative data samples or outliers, trying to assign a higher weight to dense regions of the input space, which are subject to queries [26]. *Hybrid models* attempt to combine multiple criteria for query selection, aiming at selecting data samples that are informative, i.e. close to the decision boundary of the learning model, and representative, i.e. part of a group of unlabeled data samples and unlikely to be obvious outliers.

### D. Active learning for anomaly detection

Although most active learning methods are frequently focused on image classification problems, recent literature showed an increasing interest towards anomaly detection. The authors in [27] propose an active learning strategy in the context of network intrusion detection that extends the Support Vector Domain Description method (SVDD) by querying low-confidence observations in the data. The model is retrained after querying the labels for such observations, using unlabeled data in combination with newly labeled data. A different strategy is proposed in [28], where the most likely anomalous data samples identified by unsupervised models are selected for relabelling and are used retrain a neural network-based model. A query-by-committee active learning approach proposed in [29] is adopted to let the oracle judge cases of agreement and disagreement between multiple models in a distributed and cooperative learning environment. A semi-supervised VAE-based approach is proposed in [30], where active learning is used to select a small number of samples leveraging model uncertainty via the evidence of (variational) lower bound (ELBO). An ensemble learning approach is used in [31], where multiple models are used to predict anomaly scores and the maximum anomaly score is considered as oracle's feedback. Active learning strategies in the context of credit card fraud detection are explored in [32], including standard active learning, exploratory active learning, semi-supervised learning, and their combination. Results show that simple approaches such as highest risk querying noticeably improve model's accuracy, in opposition to exploratory and unsupervised active learning techniques. Another combined modeling approach is proposed in [33], where the authors propose the interaction between a supervised and an unsupervised model. Two active strategies are proposed: the former is based on model uncertainty, while the latter select the samples for which the two models are in disagreement. An extensive experimental study with one-class learning models in active anomaly detection has been conducted in [34], highlighting the validity of both random and principled strategies.

## III. Method

In this paper, we focus on lifelong anomaly detection in a class-incremental scenario. In such scenario, given $N$ tasks, $\mathbf{T}$ is a sequence of training sets, one for each task, and $\mathbf{E}$ is a sequence of evaluation data sets corresponding to each task. Anomaly detection models are presented with one training task at a time ($T_i \in T$) and their performance is evaluated on all $E_i \in E$. Models are updated using a replay buffer $R = \{R_1 \cup R_2 \cup \ldots \cup R_N\}, R_i \subset T_i$ which contains a summarized view of representative data points in each task, and allows the model to retain knowledge about previously learned tasks while incorporating knowledge about new tasks. The number of samples stored in the memory and therefore in the replay buffer is limited. This approach is in line with replay-based lifelong learning methods [17].

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### A. General framework for mitigating anomalies from replay buffers

A crucial issue in the context of unsupervised lifelong anomaly detection is that training data sets $T_i$ may be contaminated, containing a varying rate of anomalies. Since the replay buffer $R$ is built based on the original data for each task, anomalies in the data may result in anomalies in the replay buffer, which is used to update models. As a result, this issue may have influence on the inference capabilities of anomaly detection models during the evaluation phase, i.e. models may incorporate anomalies in the learned normal data distribution. Consequently, they may tend to flag anomaly data points as normal, resulting in an increased number of false negatives. This problem is exacerbated when anomalies from multiple tasks are included in data used to update models, which represents a more complex scenario than non-lifelong counterparts typically analyzed in the literature.

Our framework takes the replay buffer $R_i$ as input and, each time a new task $T_i$ is presented, it returns a cleaned version of the replay buffer $R_i^{'}$ in which possible anomalies are removed leveraging an active learning approach. Specifically, few data points from the replay buffers are selected for oracle's labelling, according to a desired budget $B$. Subsequently, an automated strategy takes place, exploiting oracle's feedback to remove a larger number of possible anomalies without performing additional queries. In principle, using $R_i^{'}$ for model update instead of $R_i$ should yield a more robust anomaly detection model that is less subject to contamination. A graphical representation of the proposed framework is shown in Figure 1.

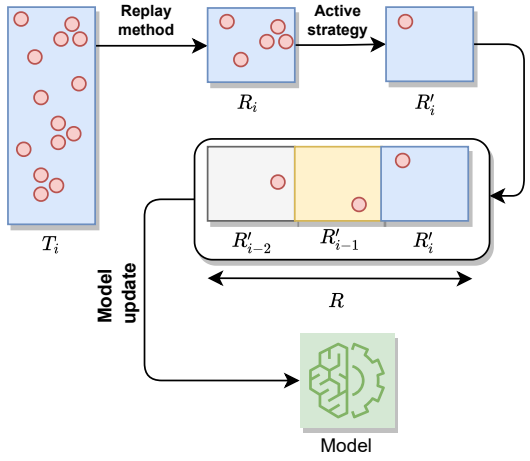



Fig. 1. Proposed framework for active lifelong anomaly detection. Data from each task $T_i$ is provided to a *replay method* that yields a compact version of the task $R_i$. The *active learning strategy* reduces the number of potential anomalies (represented as red circles) residing in the replay buffer and yields $R_i'$. All tasks are included in $R$ which is used for model update.

To accomplish this goal, we devise two distinct querying and cleaning strategies: Anomalous Cluster Removal strategy (ACR) and Similarity of Anomaly Scores strategy (SAS). While ACR analyzes the input feature space of data, SAS focuses on the analysis of anomaly scores returned by the model. Both strategies try to leverage the oracle's feedback to further clean the replay buffer, also including data points that are likely to be anomalies but were not included in oracle's queries.

### B. Anomalous Cluster Removal strategy (ACR)

In the first step, the strategy leverages the input data space to cluster data points in the replay buffer. The $k$-Means algorithm is used to partition the available data points into $k$ sets, using the original feature representation of data. The value for the $k$ parameter depends on the available number of queries (budget $B$) and it determined according to the following equation:

$$k(B) = \min(B \ \cdot 10, \ |R_i|). \tag{1}$$

In the second step, the strategy randomly selects 10% of the number of clusters $k(B)$ generated according to the budget. The rationale for the choice of this percentage value is that we are interested in reducing the probability that heterogeneous data points (normal and anomalies) are included in the same cluster. That is why it is important to set the number of clusters to a much higher value than the budget $B$. This choice is important since, considering that clusters would be fully removed in the following step if labelled as anomaly, heterogeneous data points in the same cluster would imply that normal data points are removed from $R$. Therefore, the size of each cluster should be small enough to allow for a safe removal of potential anomalies. At the same, we aim at keeping the strategy parameter-less to facilitate its adoption. Each of the randomly selected clusters of data points is a candidate for elimination from the replay buffer. The cluster is eliminated if a point $x_l$ closest to the center of the cluster has been labelled as anomaly by the oracle:

$$\begin{aligned} R_i' = R_i \setminus \{ \ x \ \mid \ & x \in R_i, \ \ c(x) = c(x_l), \\ & O(q_l) = \text{anomaly} \\ & l \in \{0, 1, \ldots B-1\}\}, \end{aligned} \tag{2}$$

where $l$ describes the id of the query sent to the oracle, $O(x_l)$ the label assigned by the oracle and $c(x)$ is a function that returns the cluster id of a data point.

Intuitively, this strategy focuses on the exploitation of diversity and similarity in data leveraging the feature representation of input data. While diversity is ensured by the clustering step, which guarantees that different data points are assigned to different clusters, similarity is exploited in the cleaning step, which removes a full cluster of potential anomalies without human intervention and without defining strict heuristics or thresholding schemes. Moreover, the elimination step is supposed to exploit the existence of micro-clusters of anomalies in the data [35]. A graphical representation of the ACR strategy is shown in Figure 2.

### C. Similarity of Anomaly Scores strategy (SAS)

Query strategies leveraging unsupervised models typically focus on selecting data points with the highest anomaly score

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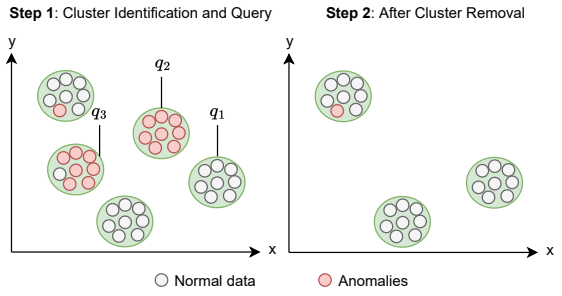



Fig. 2. ACR strategy: Step 1 is focused on clusters identification and selection of a small number of clusters ($q_1, q_2, q_3$), from each of which the data point closest to the cluster center is subject to oracle queries. Queries labelled as anomalies lead to the removal of the entire cluster they belong to.

as queries. However, this approach presents the drawback of not fully exploring the spectrum of the anomaly scores provided by the model, neglecting the fact that anomalies may also present low anomaly scores in model's predictions. In the first step, our proposed strategy queries data points based on their frequency within the full anomaly score range. We discretize the anomaly score range using an equal frequency approach [36], selecting a number of bins equal to the number of queries available. Subsequently, we select the data point to query, $q_l$, as the median data sample within each bin. More formally:

$$q_l = R_i^* \Big[ (l + 0.5) \cdot \frac{|R_i|}{B \cdot 10} \Big], \tag{3}$$

where $R_i^*$ is the list of samples in $R_i$ sorted by anomaly scores by ascending order, $R_i^*[k]$ is k-th element of list $R_i^*$ and $l = \{ 0, 1, \ldots, B-1 \}$ is an incremental index for the query.

Intuitively, this query strategy has the advantage to fully explore the anomaly score range, potentially identifying anomalies that were hidden in the lower range of anomaly scores, i.e. anomalies that the model was confidently predicting as normal. Moreover, this strategy allows us to implicitly exploit the confidence of unsupervised models without relying on an explicit measure of predictive confidence that is normally only available with supervised models such as Isolation Forest and Naive Bayes.

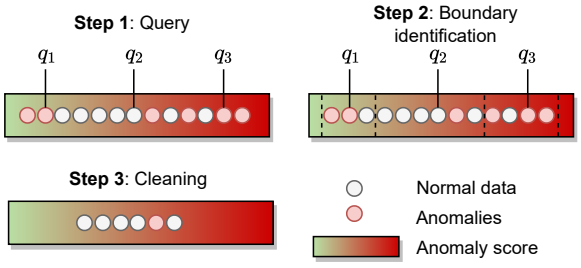



Fig. 3. SAS strategy: Step 1 queries data samples for the oracle with significantly different anomaly score values, taking into account the full range of anomaly scores returned by the model. Step 2 identifies boundaries for the neighborhood of each sample labelled as anomaly. Step 3 removes identified anomalies and their neighborhood from the replay buffer.

The second part of the strategy leverages data points labelled by the oracle and automatically labels additional data points that are close to those already labelled as anomalies. Our assumption is that, even if anomalies may present a diverse anomaly score from model's perspective, data points with a similar anomaly score of those labelled as anomalies are also likely to be anomalies, based on a locality criterion. Neighboring data points of labelled anomalies are selected to be removed, without additional oracle's intervention. More formally:

$$\begin{aligned} R_i' = R_i \setminus \{ x \mid \gamma(q_l) - \lambda \leq \gamma(x) \leq \gamma(q_l) + \lambda, \\ \lambda = \frac{b-a}{B \cdot 10}, \\ a = \min_{x \in R_i}(\gamma(x)), \quad b = \max_{x \in R_i}(\gamma(x)), \\ O(q_l) = \text{anomaly}, \\ l \in \{0, 1, \ldots, B-1\}, \}, \end{aligned} \tag{4}$$

where $\gamma$ is the anomaly score returned by the model and $O(q_l)$ the label assigned by the oracle. A graphical representation of the SAS strategy is shown in Figure 3.

## IV. Experimental Setup

Our experiments are designed to answer the following research questions:

- **RQ1:** To what extent does contamination in background data affect the anomaly detection performance of models?
- **RQ2:** Can our active lifelong anomaly detection framework boost the anomaly detection performance of models by properly removing anomalies from replay buffers?
- **RQ3:** In which scenarios do our active strategies improve model performance? Does a generalized optimal combination of active strategy and anomaly detection model exist?

To answer these questions, we perform a set of quantitative and qualitative experiments that assess the performance of models from different perspectives. In the following subsections, we describe the evaluation protocol followed in our study, including the metrics used for anomaly detection and lifelong learning, the datasets adopted and the learning scenarios addressed, as well as the competitor methods considered and their configuration. Results are discussed in Section V.

### A. Evaluation protocol and metrics

We follow the evaluation protocol applied in most lifelong learning works [37] and described in Section II. One important aspect of the protocol is having access to training $T$ and testing $E$ sets for all tasks. The scenarios considered in this study are class-incremental, which means that the model is provided with information about the transition between tasks, but has no indication about which task is currently processed. In our experiments, all models are are presented with a sequence of training sets $T$ and are updated after introducing each task $T_i$. After learning task $T_i$, the model is evaluated on testing sets from all tasks processed so far $E_j$, $j \leq i$. The scenario described so far is represented in Figure 4. This approach allows us to determine the performance of models

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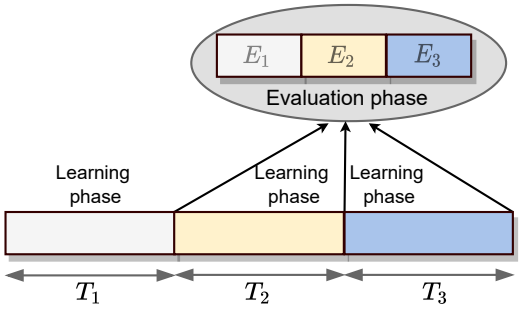



Fig. 4. Example of training sets representing a changing data distribution with tasks $T_1, T_2, T_3$ and corresponding testing sets $E_1, E_2, E_3$.

on previously learned tasks and therefore assess their ability to retain knowledge.

Our evaluation protocol produces a matrix $R_{N\times N}$, $N = |T|$ that gathers all obtained results. $R_{i,j}$ represents the ROC-AUC metric of the model evaluated on task $j$ after learning task $i$. We showcase this process in Algorithm 1. The matrix $R_{N\times N}$ is then leveraged by us to compute the lifelong learning metrics.

**Input:** **T** – a sequence of $N$ training sets
**Input:** **E** – a sequence of $N$ testing sets
**Input:** $L$ – learning model
Initialize results matrix $R_{N\times N}$
**for** $T_i \in$ **T** **do**
  Train $L$ on $T_i$
  **for** $j \in \{0, 1, \ldots, i\}$ **do**
    Evaluate model $L$ on data $E_j$ and compute ROC-AUC
    Save results to $R_{i,j}$
  **end**
**end**
**return** $R$

**Algorithm 1:** Pseudo-code of the evaluation protocol

The most widely adopted metric in lifelong learning image classification is accuracy. However, this metric is not well-suited for anomaly detection due to the inherent differences in terms of problem definition and evaluation requirements among the two problems. One of the best suited metrics for assessing anomaly detection performance is ROC-AUC (Area Under Receiver Operating Characteristic Curve). A ROC plot shows the trade-off between the true negative rate and true positive rate. The essential advantage of ROC-AUC is being a threshold-free measure, which does not require a hard-set threshold and allows to assess the capabilities of the model in a much more complete way than threshold dependent metrics. In this work, we extend the definition of ROC-AUC to its lifelong version, so that it can properly measure the performance of the models on not only one task, but on all previously learned tasks after learning each new task. This approach conforms to the the definition of lifelong accuracy proposed in [37] and is defined as in the following:

$$\text{Lifelong ROC-AUC} = \frac{\sum_{i\geq j}^{N} R_{i,j}}{\frac{N(N-1)}{2}} \tag{5}$$

### B. Datasets and scenarios

We perform experiments leveraging the following datasets:

- **NSL-KDD** [38] is a network-based intrusion dataset containing records of network traffic from the DARPA Intrusion Detection Systems program. It was built upon KDD-99 to solve the most important issues of its predecessor;
- **UNSW-NB15** [39] is a network-based intrusion dataset containing a mixture of modern network traffic from real modern setup and from synthesized malicious activities;
- **NGIDS-IDS** [40] is a host-based anomaly detection dataset created with modern security challenges in mind. It contains attacks from seven different families.

We create a few class-incremental lifelong learning scenarios based on the described datasets adopting the following procedure:

1) We cluster normal data into the desired number of clusters (reflecting the number of tasks $N$) leveraging a centroid-based clustering algorithm such as $k$-means.
2) We also cluster anomalies into the desired number of clusters.
3) We merge one cluster of normal class with its closest cluster of anomalies into a single task.
4) We separate each task into training and testing sets. The training set contains the specified number of anomalies according to a desired contamination rate $C$.

This procedure allows us to create class-incremental lifelong learning scenarios containing a specified number of tasks and contamination rate in the training set.

### C. Methods

In our experiments, we use four unsupervised and semi-supervised anomaly detection models:

- **AE** [41]: Autoencoders are very popular neural network models that learn data distributions in a one-class manner by minimizing reconstruction error, which is defined as the difference between original and reconstructed data. The reconstruction error can be also used as an anomaly score, using the assumption that data from the originally learned distribution will present much lower reconstruction error than anomalies.
- **One Class Support Vector Machine (OC-SVM)** [42]: OC-SVM is a kernel-based semi-supervised method that aims to learn a separating hyperplane in a high-dimensional space. OC-SVM provides anomaly scores based on its comparison to the training distribution and its position inside the decision boundary.
- **Local Outlier Factor (LOF)** [43]: LOF is a method measuring the local density deviation of data with respect to its neighbors. The anomaly score is provided based on the ratio between the local density of the sample and the average local density of the nearest neighbors.

We also compare the two proposed ACR and SAS strategies with two commonly adopted baseline query strategies applied in non-lifelong active anomaly detection works:

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- **Random (RND)** – this simple query strategy randomly selects $B$ samples that are later labelled by an oracle and removed if they are truly anomalies. Despite being a very simple and completely random-based method, recent studies showed that it may yield promising results that outperform more complex strategies [34].
- **Highest Anomaly Score (HAS)** – this query strategy follows the assumption that the anomalous samples will have the highest anomaly score. In order to generate anomaly scores, the anomaly detection model (one of the four used in our study) is used in an unsupervised manner each time the strategy is applied. Once anomaly scores are predicted, the $B$ samples with the highest value of anomaly score are labelled by the oracle and removed if they are true anomalies [32].

In our experiments, we adopt a state-of-the-art combination of balanced replay buffers and reservoir sampling [17]. The space in the replay memory is equally divided between each task $T_i$, while decision which samples from the specific task $T_i$ should be kept is made leveraging reservoir sampling [18]. This approach ensures the proper balance between each task.

## V. Results and Discussion

Results in Tables I, II, and III show the anomaly detection performance in terms of Lifelong ROC-AUC for all models (LOF, OC-SVM, AE) and strategies (Rand, HAS, ACR, SAS) with varying contamination rates ($C$) and budget levels ($B$). In the following subsections, we comments results in light of our research questions separately.

### *A. Impact of contamination on anomaly detection performance.*

In this subsection our goal is to measure and analyze the impact of contamination on the anomaly detection performance of models in a class-incremental lifelong anomaly detection setting. Moreover, we showcase how the performance of different models can be improved by cleaning replay buffers from anomalies.

The results in Figure 5 show the performance of all models at varying contamination rates and assuming the ability to remove a different percentages of anomalies. Results are extracted adopting a budget-free variant of the HAS strategy, where the oracle receives queries until a desired amount of anomalies is removed $(20\%, 40\%, 60\%, 80\%, 100\%)$.

Results highlight some interesting patterns concerning the impact of data contamination on models' performance. Specifically, LOF has the potential to yield a high performance on all datasets, but its inferences are severely affected by contamination. As a result, its performance remains low when a moderate quantity of anomalies is removed $(20\% - 60\%)$, and it is strongly improved when the majority of the anomalies are removed $(80\% - 100\%)$.

Despite the fact that OC-SVM presents a poor performance on NGIDS independently from how many anomalies are present in the data, it is the most stable method with respect to contamination rates considering the other datasets. Indeed,

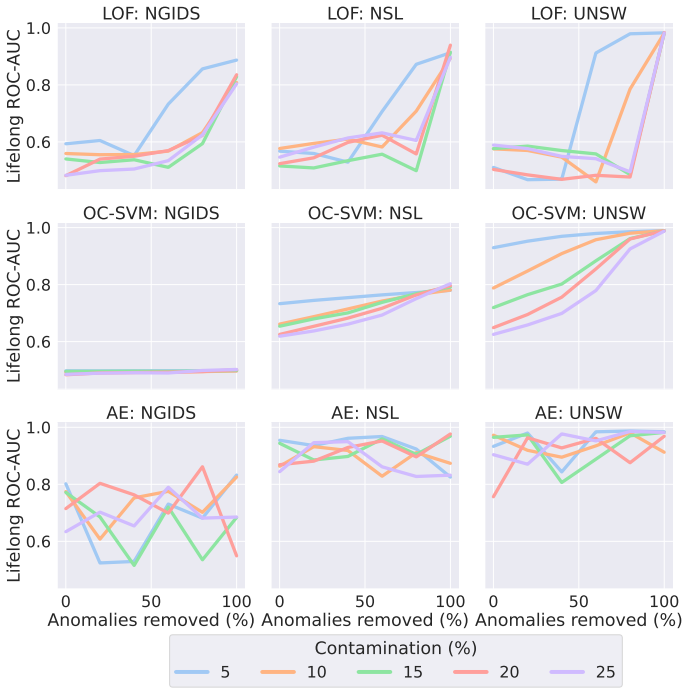



Fig. 5. Experimental study of the impact of contamination and replay buffer cleaning capabilities on the anomaly detection performance of all methods (LOF, OC-SVM, AE) and datasets (NGIDS, NSL, UNSW). Performance is measured in terms of Lifelong ROC-AUC with varying contamination rates (5%-25%) and simulating the removal of different percentages of anomalies from the replay buffer.

as more anomalies are removed, the performance of the model gradually improves, which denotes that the ability to remove any amount of anomalies is beneficial for the model.

AE generally exhibits a better performance when compared to LOF and OC-SVM, but its trend is less consistent than other methods with different percentages of removed anomalies. While on NSL and UNSW a non-monotonic increasing trend could still be observed, this does not occur with NGIDS, where AE's performance oscillates without a clear trend. This behavior is possibly due to the stochastic optimization of the model and the limited replay buffer size available for this dataset (500 samples per task).

Overall, our analysis show that contamination has a significant impact on the performance of anomaly detection models, and suggests that active learning strategies in the context of lifelong anomaly detection present a great potential to yield more robust models. Removing a sufficient amount of anomalies is positively correlated with model's performance to a varying degree based on model's characteristics. In some cases, removing a large number of anomalies is fundamental to make models successful, which suggests the importance of active learning strategies that adopt automated exploration and exploitation mechanisms in addition to oracle's queries (**RQ1**). However, they yield marginal improvements if the base models are unable to properly fit the characteristics of a specific dataset, as seen for OC-SVM with the NGIDS dataset.

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TABLE I

EXPERIMENTAL RESULTS FOR THE NGIDS DATASET. LIFELONG ROC-AUC FOR ALL MODELS (LOF, OC-SVM, AE), CONTAMINATION RATES $C$, BUDGETS $B$, AND STRATEGIES (RND, HAS, ACR, SAS). THE BEST RESULTS FOR EACH COMBINATION OF METHOD, DATASET, BUDGET, AND CONTAMINATION RATE ARE SHOWN IN BOLD.

| C | B | LOF | | | | OC-SVM | | | | AE | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RND | HAS | ACR | SAS | RND | HAS | ACR | SAS | RND | HAS | ACR | SAS |
| 0.05 | 0 | 0.593 | | | | 0.497 | | | | 0.615 | | | |
| | 5 | 0.530 | 0.593 | 0.591 | **0.830** | **0.504** | 0.502 | 0.492 | 0.501 | 0.595 | 0.612 | **0.698** | 0.566 |
| | 15 | 0.575 | 0.593 | 0.570 | **0.831** | 0.498 | **0.502** | 0.495 | 0.501 | 0.689 | 0.574 | 0.549 | **0.723** |
| | 25 | 0.560 | 0.593 | 0.546 | **0.884** | 0.503 | **0.504** | 0.497 | 0.503 | 0.552 | 0.701 | **0.770** | 0.620 |
| | 50 | 0.536 | 0.598 | 0.544 | **0.887** | 0.486 | **0.505** | 0.486 | 0.503 | 0.559 | 0.555 | 0.521 | **0.782** |
| | 100 | 0.547 | 0.609 | 0.546 | **0.887** | 0.501 | **0.505** | 0.502 | 0.503 | 0.469 | 0.555 | **0.782** | 0.688 |
| 0.10 | 0 | 0.559 | | | | 0.493 | | | | 0.707 | | | |
| | 5 | 0.524 | 0.559 | 0.503 | **0.603** | **0.504** | 0.492 | 0.489 | 0.492 | 0.513 | 0.540 | 0.366 | **0.734** |
| | 15 | 0.518 | 0.560 | 0.553 | **0.785** | 0.507 | 0.492 | **0.508** | 0.491 | 0.718 | 0.733 | 0.710 | **0.785** |
| | 25 | 0.558 | 0.561 | 0.557 | **0.801** | **0.500** | 0.498 | 0.498 | 0.491 | **0.698** | 0.546 | 0.389 | 0.451 |
| | 50 | 0.532 | 0.555 | 0.718 | **0.806** | 0.484 | **0.501** | 0.485 | 0.491 | 0.500 | 0.513 | **0.683** | 0.615 |
| | 100 | 0.521 | 0.576 | **0.686** | 0.805 | 0.478 | **0.501** | 0.481 | 0.491 | 0.470 | 0.556 | 0.755 | **0.811** |
| 0.15 | 0 | 0.540 | | | | 0.496 | | | | 0.530 | | | |
| | 5 | 0.516 | 0.540 | 0.561 | **0.774** | **0.495** | 0.492 | 0.482 | 0.492 | 0.712 | **0.763** | 0.535 | 0.555 |
| | 15 | 0.486 | 0.538 | 0.529 | **0.819** | **0.495** | 0.492 | 0.480 | 0.492 | 0.530 | 0.515 | 0.525 | **0.639** |
| | 25 | 0.507 | 0.540 | 0.494 | **0.825** | 0.490 | **0.495** | 0.489 | 0.491 | **0.674** | 0.559 | 0.532 | 0.618 |
| | 50 | 0.530 | 0.542 | 0.550 | **0.825** | 0.494 | **0.498** | 0.494 | 0.493 | 0.621 | 0.479 | 0.377 | **0.698** |
| | 100 | 0.521 | 0.513 | 0.553 | **0.829** | 0.487 | **0.502** | 0.487 | 0.499 | 0.529 | 0.553 | 0.561 | **0.746** |
| 0.20 | 0 | 0.482 | | | | 0.484 | | | | 0.475 | | | |
| | 5 | 0.471 | 0.481 | 0.485 | **0.805** | 0.495 | **0.498** | 0.492 | **0.498** | **0.720** | 0.632 | 0.532 | 0.613 |
| | 15 | 0.528 | 0.484 | 0.505 | **0.834** | 0.497 | 0.498 | 0.494 | **0.499** | 0.605 | 0.714 | **0.785** | 0.540 |
| | 25 | 0.517 | 0.486 | 0.657 | **0.834** | 0.496 | **0.500** | 0.488 | 0.499 | 0.443 | 0.411 | 0.531 | **0.542** |
| | 50 | 0.522 | 0.485 | 0.527 | **0.834** | 0.498 | **0.504** | 0.492 | 0.496 | **0.604** | 0.585 | 0.545 | 0.548 |
| | 100 | 0.543 | 0.507 | 0.539 | **0.834** | 0.486 | **0.512** | 0.485 | 0.498 | 0.496 | 0.519 | 0.352 | **0.545** |
| 0.25 | 0 | 0.483 | | | | 0.485 | | | | 0.447 | | | |
| | 5 | 0.504 | 0.483 | 0.536 | **0.767** | 0.485 | 0.484 | **0.486** | 0.482 | 0.446 | 0.512 | **0.670** | 0.550 |
| | 15 | 0.459 | 0.483 | 0.483 | **0.796** | 0.481 | 0.484 | **0.497** | 0.485 | 0.480 | 0.579 | 0.438 | **0.587** |
| | 25 | 0.457 | 0.488 | 0.482 | **0.796** | 0.483 | **0.491** | 0.484 | 0.487 | **0.723** | 0.510 | 0.645 | 0.647 |
| | 50 | 0.514 | 0.490 | 0.509 | **0.796** | 0.482 | **0.495** | 0.481 | 0.486 | 0.510 | 0.457 | 0.599 | **0.715** |
| | 100 | 0.484 | 0.485 | 0.677 | **0.796** | 0.482 | **0.504** | 0.480 | 0.499 | 0.479 | **0.835** | 0.399 | 0.576 |

TABLE II

EXPERIMENTAL RESULTS FOR THE NSL DATASET. LIFELONG ROC-AUC FOR ALL MODELS (LOF, OC-SVM, AE), CONTAMINATION RATES $C$, BUDGETS $B$ AND STRATEGIES (RND, HAS, ACR, SAS). THE BEST RESULTS FOR EACH COMBINATION OF METHOD, DATASET, BUDGET, AND CONTAMINATION RATE ARE SHOWN IN BOLD.

| C | B | LOF | | | | OC-SVM | | | | AE | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RND | HAS | ACR | SAS | RND | HAS | ACR | SAS | RND | HAS | ACR | SAS |
| 0.05 | 0 | 0.568 | | | | 0.733 | | | | 0.792 | | | |
| | 5 | **0.604** | 0.569 | 0.554 | 0.568 | 0.739 | 0.748 | **0.749** | 0.743 | 0.818 | 0.944 | **0.947** | 0.868 |
| | 15 | 0.547 | 0.570 | 0.569 | **0.635** | 0.735 | **0.759** | 0.740 | 0.743 | 0.818 | 0.831 | 0.835 | **0.960** |
| | 25 | 0.594 | 0.573 | 0.528 | **0.666** | 0.735 | **0.766** | 0.735 | 0.744 | 0.833 | **0.928** | 0.868 | 0.862 |
| | 50 | 0.587 | 0.582 | 0.588 | **0.621** | 0.740 | **0.775** | 0.739 | 0.747 | 0.857 | 0.825 | **0.947** | 0.824 |
| | 100 | 0.526 | 0.585 | 0.541 | **0.644** | 0.745 | **0.781** | 0.746 | 0.752 | 0.873 | **0.931** | 0.827 | 0.890 |
| 0.10 | 0 | 0.577 | | | | 0.661 | | | | 0.907 | | | |
| | 5 | 0.578 | 0.577 | 0.599 | **0.684** | 0.670 | **0.684** | 0.668 | 0.682 | 0.906 | **0.942** | 0.938 | 0.851 |
| | 15 | 0.551 | 0.578 | **0.588** | 0.587 | 0.673 | **0.704** | 0.672 | 0.687 | 0.852 | 0.831 | **0.945** | 0.828 |
| | 25 | **0.586** | 0.578 | 0.556 | 0.552 | 0.675 | **0.717** | 0.671 | 0.691 | 0.850 | 0.817 | 0.854 | **0.886** |
| | 50 | 0.621 | 0.588 | **0.639** | 0.575 | 0.682 | **0.731** | 0.683 | 0.693 | 0.862 | 0.925 | 0.921 | **0.951** |
| | 100 | 0.622 | 0.602 | **0.626** | 0.584 | 0.691 | **0.760** | 0.697 | 0.705 | 0.854 | **0.956** | 0.870 | 0.793 |
| 0.15 | 0 | 0.516 | | | | 0.654 | | | | 0.843 | | | |
| | 5 | 0.519 | 0.517 | 0.521 | **0.642** | 0.641 | 0.657 | 0.656 | **0.666** | 0.870 | 0.926 | **0.928** | 0.853 |
| | 15 | 0.500 | 0.518 | 0.551 | **0.606** | 0.659 | **0.675** | 0.659 | 0.668 | **0.856** | 0.821 | 0.796 | 0.855 |
| | 25 | 0.550 | 0.518 | 0.542 | **0.688** | 0.658 | **0.683** | 0.670 | 0.665 | **0.934** | 0.827 | 0.890 | 0.860 |
| | 50 | 0.449 | 0.520 | 0.483 | **0.641** | 0.653 | **0.705** | 0.656 | 0.669 | 0.924 | 0.857 | 0.934 | **0.941** |
| | 100 | 0.501 | 0.536 | 0.511 | **0.622** | 0.673 | **0.744** | 0.674 | 0.683 | 0.797 | **0.964** | 0.854 | 0.893 |
| 0.20 | 0 | 0.524 | | | | 0.625 | | | | 0.926 | | | |
| | 5 | 0.515 | 0.524 | **0.527** | 0.505 | 0.621 | 0.630 | 0.628 | **0.636** | **0.913** | 0.888 | 0.797 | 0.847 |
| | 15 | 0.545 | 0.523 | **0.556** | 0.524 | 0.628 | 0.634 | 0.639 | **0.640** | 0.844 | 0.901 | 0.857 | **0.918** |
| | 25 | 0.524 | 0.525 | 0.535 | **0.541** | 0.626 | 0.641 | 0.639 | **0.643** | 0.916 | 0.899 | **0.940** | 0.885 |
| | 50 | 0.532 | 0.527 | 0.514 | **0.545** | 0.634 | **0.668** | 0.639 | 0.653 | 0.918 | **0.936** | 0.867 | 0.917 |
| | 100 | 0.565 | 0.544 | 0.564 | **0.606** | 0.657 | **0.695** | 0.655 | 0.662 | 0.919 | **0.937** | 0.924 | 0.911 |
| 0.25 | 0 | 0.546 | | | | 0.619 | | | | 0.912 | | | |
| | 5 | 0.511 | 0.548 | 0.536 | **0.693** | 0.626 | 0.628 | **0.644** | **0.644** | 0.920 | **0.944** | 0.844 | 0.849 |
| | 15 | 0.528 | 0.551 | 0.551 | **0.669** | 0.627 | 0.632 | 0.634 | **0.643** | 0.830 | 0.858 | 0.850 | **0.928** |
| | 25 | 0.516 | 0.552 | 0.513 | **0.720** | 0.626 | **0.634** | 0.627 | 0.629 | 0.908 | 0.919 | 0.847 | **0.925** |
| | 50 | 0.531 | 0.553 | 0.538 | **0.557** | 0.633 | 0.654 | 0.629 | **0.653** | **0.903** | 0.836 | 0.897 | 0.819 |
| | 100 | 0.552 | 0.580 | 0.539 | **0.656** | 0.641 | **0.684** | 0.634 | 0.653 | **0.921** | 0.916 | 0.847 | 0.853 |

### *B. Impact of the framework on anomaly detection performance*

Starting with the NGIDS dataset, results in Table I show that OC-SVM has overall a very poor performance at varying contamination rates, even though some improvements can be observed with any active learning strategy. For example, the ACR strategy improves the anomaly detection performance in terms of Lifelong ROC-AUC from $0.493$ to $0.508$ in the scenario with $C = 0.10$ and $B = 15$. AE presents a slightly better overall performance than OC-SVM, although it is still not satisfactory without any active learning strategy. However, a larger margin of improvement can be observed for all active strategies. For instance, for $C = 0.15$ and $B = 5$, the HAS strategy improves the performance from $0.530$ to $0.763$, which is a remarkable gain. However, the outcome of the different strategies is quite unstable since a clear trend with increasing levels of budget cannot be clearly identified. LOF presents the highest overall performance on this dataset, where active strategies show their best potential. For example, our SAS strategy with a very small budget of $B = 5$ allows to improve the performance from $0.484$ to $0.805$ in the scenario with a large contamination rate $C = 0.20$. Overall, one or more active strategies appear effective in improving the anomaly detection performance in 70 out of 75 combinations of models, contamination rates, and budgets with the NGIDS dataset.

On the NSL dataset, results in Table II show that our framework provides a large improvement across all methods and contamination rates. For example, with LOF, our SAS strategy with a budget $B = 25$ yields a strong improvement, increasing model's performance from $0.546$ to $0.720$ at a high contamination rate $C = 0.25$. Considering a representative example for OC-SVM, in the scenario with contamination rate $C = 0.15$, the HAS active strategy improves its performance from $0.654$ to $0.744$ using a moderate budget of $B = 100$. For AE, a remarkable case is that of contamination rate $C = 0.05$, where active strategies yield a considerable improvement under a minimal budget. For instance, ACS improves the performance from $0.792$ to $0.947$ with $B = 5$, whereas the SAS strategy improves it to $0.960$ with $B = 15$. Overall, one or more active strategies appear effective in improving the anomaly detection performance in 71 out of 75 combinations of models, contamination rates, and budgets with the NSL dataset.

Shifting the focus on the UNSW dataset, results in Table III show that active strategies provide, in the average case, a moderately low improvement for LOF, and a more significant improvement for OC-SVM and AE. For example, the HAS strategy for OC-SVM allows to improve its performance from $0.788$ to $0.943$ and from $0.719$ to $0.842$ with a low budget $B = 25$ in the scenarios with contamination rate $C = 0.10$

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TABLE III
EXPERIMENTAL RESULTS FOR THE UNSW DATASET. LIFELONG ROC-AUC FOR ALL MODELS (LOF, OC-SVM, AE), CONTAMINATION RATES $C$, BUDGETS $B$ AND STRATEGIES (RND, HAS, ACR, SAS). THE BEST RESULTS FOR EACH COMBINATION OF METHOD, DATASET, BUDGET AND CONTAMINATION RATE ARE SHOWN IN BOLD.

| C | B | LOF | | | | OC-SVM | | | | AE | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RND | HAS | ACR | SAS | RND | HAS | ACR | SAS | RND | HAS | ACR | SAS |
| 0.05 | 0 | 0.511 | | | | 0.929 | | | | 0.893 | | | |
| | 5 | **0.531** | 0.513 | 0.513 | 0.511 | 0.943 | **0.971** | 0.950 | 0.947 | 0.931 | **0.984** | 0.919 | 0.891 |
| | 15 | **0.580** | 0.515 | 0.516 | 0.546 | 0.944 | **0.988** | 0.944 | 0.947 | 0.899 | **0.985** | 0.921 | 0.895 |
| | 25 | 0.505 | 0.516 | 0.476 | **0.592** | 0.943 | **0.989** | 0.950 | 0.947 | 0.905 | **0.986** | 0.907 | 0.879 |
| | 50 | 0.520 | **0.526** | 0.502 | 0.500 | 0.945 | **0.990** | 0.947 | 0.953 | **0.983** | 0.859 | 0.822 | 0.826 |
| | 100 | 0.511 | **0.535** | 0.488 | 0.505 | 0.959 | **0.990** | 0.960 | 0.960 | 0.892 | **0.927** | 0.895 | 0.910 |
| 0.10 | 0 | 0.576 | | | | 0.788 | | | | 0.893 | | | |
| | 5 | **0.577** | **0.577** | 0.532 | **0.577** | 0.798 | **0.834** | 0.803 | 0.792 | **0.978** | 0.888 | 0.841 | 0.814 |
| | 15 | **0.607** | 0.581 | 0.558 | 0.575 | 0.812 | **0.912** | 0.812 | 0.799 | 0.879 | 0.890 | **0.965** | 0.788 |
| | 25 | 0.576 | 0.583 | **0.608** | 0.568 | 0.809 | **0.943** | 0.815 | 0.795 | 0.911 | 0.933 | 0.916 | **0.978** |
| | 50 | 0.557 | **0.598** | 0.560 | 0.548 | 0.821 | **0.958** | 0.827 | 0.805 | **0.981** | 0.910 | 0.895 | 0.890 |
| | 100 | 0.558 | **0.613** | 0.565 | 0.582 | 0.847 | **0.971** | 0.852 | 0.824 | 0.979 | 0.915 | **0.980** | 0.919 |
| 0.15 | 0 | 0.577 | | | | 0.719 | | | | 0.860 | | | |
| | 5 | 0.599 | 0.577 | 0.571 | **0.584** | 0.718 | **0.756** | 0.748 | 0.726 | **0.971** | 0.880 | 0.889 | 0.879 |
| | 15 | 0.561 | 0.577 | 0.601 | **0.612** | 0.743 | **0.805** | 0.753 | 0.730 | 0.829 | **0.890** | 0.870 | 0.768 |
| | 25 | 0.568 | **0.582** | 0.572 | 0.580 | 0.733 | **0.842** | 0.742 | 0.734 | 0.877 | 0.867 | 0.813 | **0.885** |
| | 50 | **0.608** | 0.590 | 0.601 | 0.564 | 0.741 | **0.862** | 0.738 | 0.737 | 0.881 | **0.897** | 0.887 | 0.747 |
| | 100 | 0.577 | **0.628** | 0.584 | 0.560 | 0.762 | **0.906** | 0.772 | 0.747 | 0.887 | 0.910 | **0.976** | 0.875 |
| 0.20 | 0 | 0.503 | | | | 0.649 | | | | 0.860 | | | |
| | 5 | 0.521 | 0.504 | **0.547** | 0.503 | 0.651 | **0.676** | 0.671 | 0.645 | 0.787 | 0.816 | **0.864** | 0.849 |
| | 15 | 0.470 | 0.506 | **0.552** | 0.514 | 0.651 | **0.732** | 0.658 | 0.646 | 0.815 | **0.889** | 0.733 | 0.736 |
| | 25 | 0.524 | 0.509 | **0.542** | 0.509 | 0.682 | **0.762** | 0.702 | 0.650 | 0.796 | 0.892 | 0.722 | **0.966** |
| | 50 | 0.512 | **0.513** | 0.509 | 0.489 | 0.653 | **0.789** | 0.670 | 0.663 | 0.846 | 0.783 | **0.968** | 0.846 |
| | 100 | 0.519 | **0.538** | 0.537 | 0.477 | 0.699 | **0.821** | 0.706 | 0.679 | **0.971** | 0.917 | 0.844 | 0.789 |
| 0.25 | 0 | 0.589 | | | | 0.625 | | | | 0.739 | | | |
| | 5 | 0.525 | **0.589** | 0.569 | 0.587 | 0.619 | **0.655** | 0.623 | 0.631 | **0.958** | 0.801 | **0.958** | 0.790 |
| | 15 | 0.522 | **0.592** | 0.545 | 0.575 | 0.620 | **0.707** | 0.660 | 0.626 | 0.846 | 0.833 | **0.963** | 0.806 |
| | 25 | 0.555 | **0.594** | 0.554 | 0.547 | 0.636 | **0.727** | 0.637 | 0.623 | 0.808 | 0.814 | **0.959** | 0.725 |
| | 50 | 0.557 | **0.606** | 0.547 | 0.531 | 0.650 | **0.751** | 0.637 | 0.631 | 0.961 | 0.858 | 0.867 | **0.963** |
| | 100 | 0.570 | **0.629** | 0.532 | 0.532 | 0.674 | **0.777** | 0.647 | 0.658 | 0.725 | **0.909** | 0.779 | 0.882 |

and $C = 0.15$, respectively. With the same budget, results with AE and the SAS strategy show an improvement from $0.893$ to $0.978$ and from $0.860$ to $0.966$ with contamination rate $C = 0.10$ and $C = 0.20$, respectively. The highest performance is achieved by the ACR strategy with a budget $B = 100$ and contamination rate $C = 0.10$, increasing model's performance from $0.893$ to $0.980$. Overall, one or more active strategies show to be effective in improving the anomaly detection performance in all 75 combinations of models, contamination rates, and budgets with the UNSW dataset.

In general, our experiments clearly show that our proposed active learning framework allows to improve the performance of models in the lifelong anomaly detection problem (**RQ2**). However, a moderate to large variability in the improvements can be observed across different datasets, as well as different contamination rates and budgets, and deserves further investigation.

### C. Interaction between different strategies and methods

We analyze the performance of different strategies, and compare baseline strategies (RND and HAS) with our two proposed strategies (ACR and SAS).

Starting with the NGIDS dataset, results in Table I show that OC-SVM yields a poor performance, showcasing the inability to learn a proper separation boundary, which makes drawing considerations about the effectiveness of active strategies very challenging. On the other hand, the SAS strategy strongly improves the performance for LOF in all budget and contamination configurations, significantly outperforming other strategies. For AE, the situation is more uneven, and it is difficult to highlight a clear winning active strategy. However, our SAS strategy improves baseline strategies by a significant margin in the majority of the cases (15 out of 25), whereas ACR improves baselines in 10 out of 25 cases.

On the NSL dataset, results in Table II show that the HAS strategy is the best for OC-SVM (19 out of 25 cases), and the largest margin is achieved with higher budget values. However, all active strategies yield a similar performance gain. For LOF, it is difficult to highlight the best strategy. However SAS improves baselines in the majority of the cases (19 out of 25), whereas ACR improves baselines in 10 out of 25 cases, both with a varying margin of performance gain. For AE, baseline strategies are the best in 12 cases out of 25, whereas our proposed strategies yield the best performance in 13 out of 25 cases.

Shifting the focus on the UNSW dataset, the HAS strategy surprisingly yields the highest performance for OC-SVM in all 25 configurations, indicating that the anomaly scores provided by the model on data in the replay buffer represent a reliable way to isolate anomalies from normal data in this dataset. LOF appears to yield a very weak performance on this dataset, and all active strategies appear to be insufficient to improve its inferences. For AE, at low contamination rates, HAS presents a very good performance. On the other hand, for high contamination rates and low budgets, ACR is the preferred strategy. For example, with $C = 0.25$ and $B = 15$, ACR allows to boost the performance from $0.739$ to $0.963$.

Overall, taking into consideration the datasets where the base models can achieve reasonable results, OC-SVM seems to integrate better with simpler strategies such as HAS, whereas LOF yields a significantly stronger performance with the SAS strategy. AE presents a very good ability to successfully incorporate all active strategies to boost anomaly detection performance (**RQ3**).

## VI. CONCLUSION

In this work, we explored active learning in the context of the lifelong anomaly detection problem. We proposed a framework for class-incremental scenarios that supports any memory-based experience replay method and any query strategy. Leveraging experience replay, the framework allows models to consolidate past knowledge while they adapt to new knowledge. Our active learning module further enhances this capability reducing the number of anomalies in the replay buffer. We performed experiments on three network and host-based intrusion detection datasets and involving four active learning strategies. Results showed that active lifelong anomaly detection can be a valuable asset to increase the longevity and robustness of models, and that our proposed framework is able to improve the anomaly detection performance of different models under low labeling budget con-

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straints. As future work, we aim to investigate how active learning anomaly detection impacts model forgetting on previously learned tasks. Additional research directions include the adoption of curriculum learning and multi-dataset learning capabilities in lifelong anomaly detection models.

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